

Course on Probabilistic Design of Coastal Structures

Statistical Modelling of Environmental Conditions

02. ANNUAL WEATHER TYPES (AWT)

Jared Ortiz-Angulo Cantos

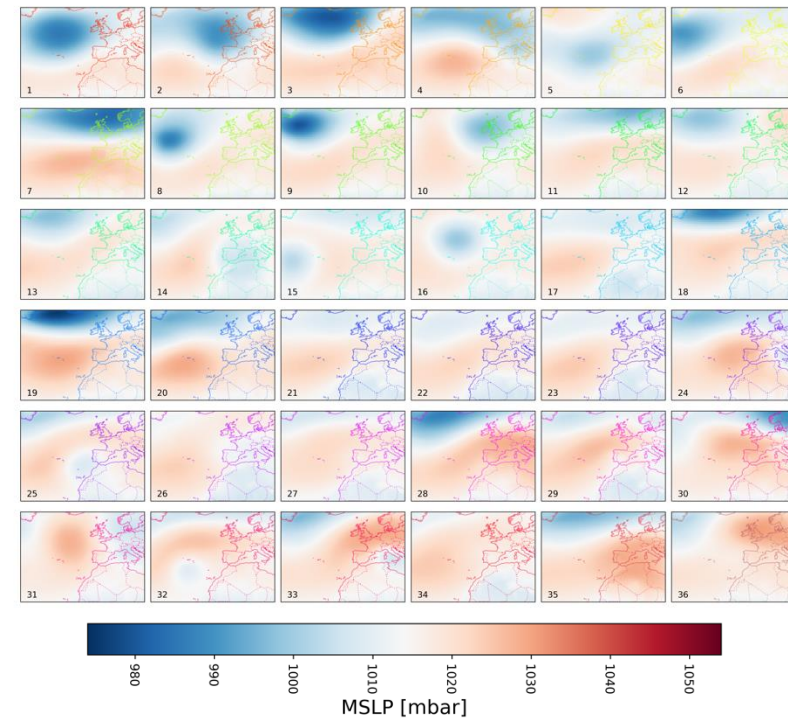
Researcher, GeoOcean Group, Universidad de Cantabria (UC)

ProCoast 2025
October 20th-23th, 2025

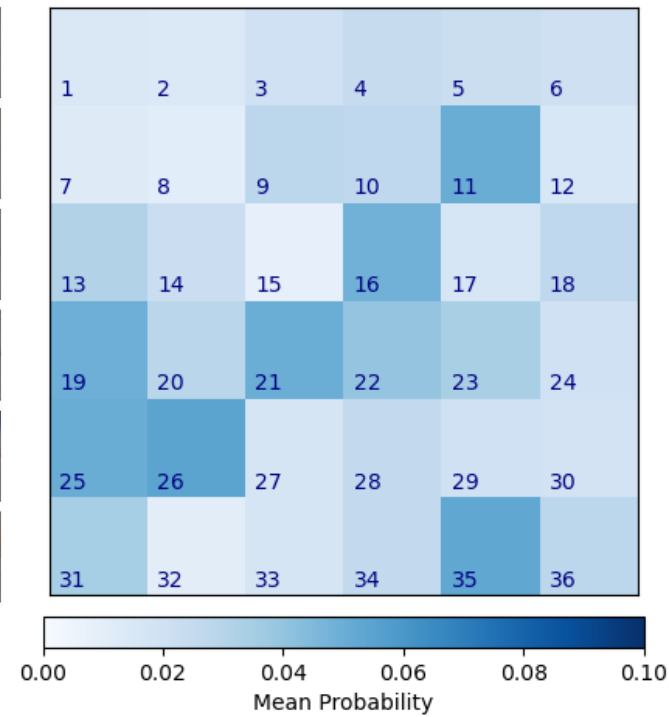


Framework

DAILY WEATHER TYPES (DWT)



MEAN PROBABILITY OF THE DWT (1944-2023)



ANNUAL PROBABILITY OF THE DWT



Framework

ANNUAL PROBABILITY OF THE DWT

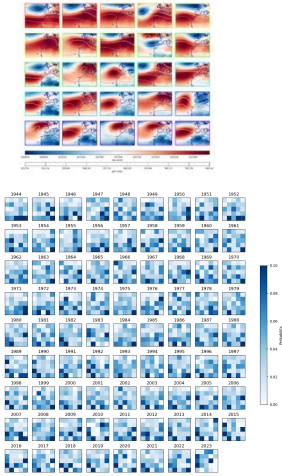


Interannuality (AWT)

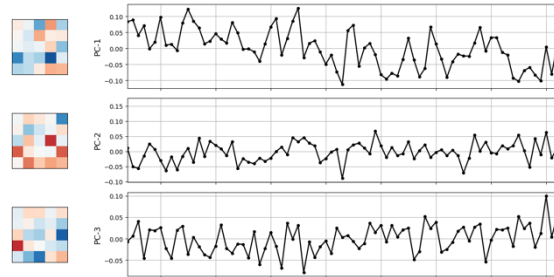
Framework

1. DWTs

1944
1953
ANNUAL PROBABILITY ASSOCIATED WITH THE DWTs

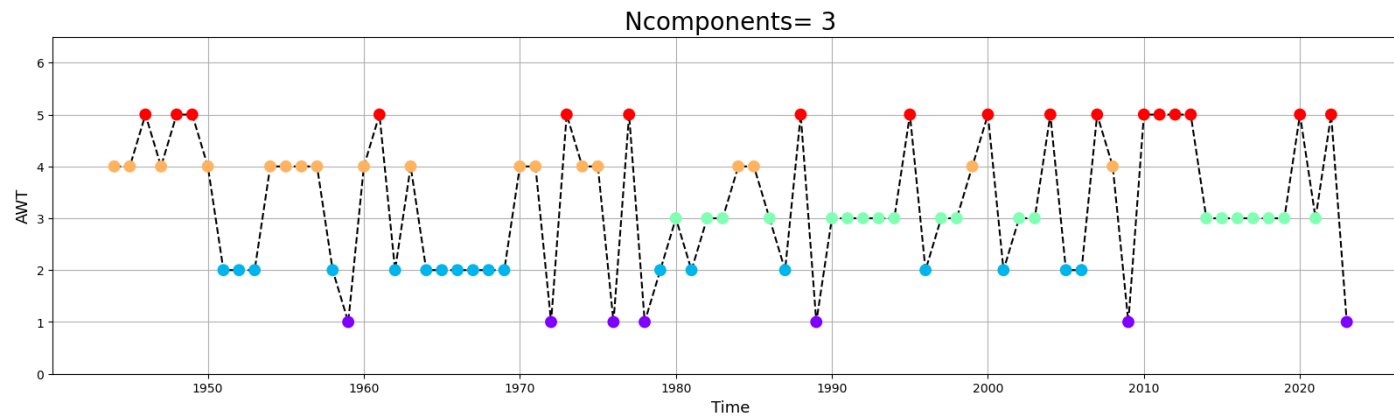
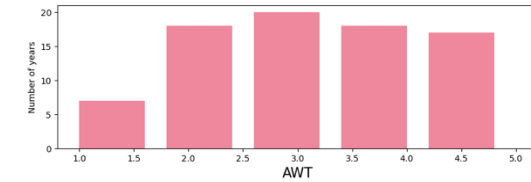


2. PCA



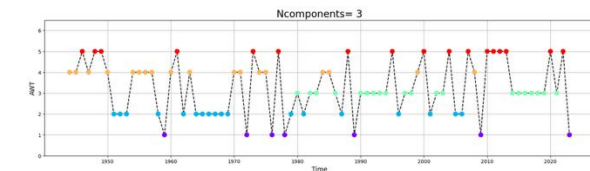
3. KMeans

Nº PCs = 3
Nº clusters = 5

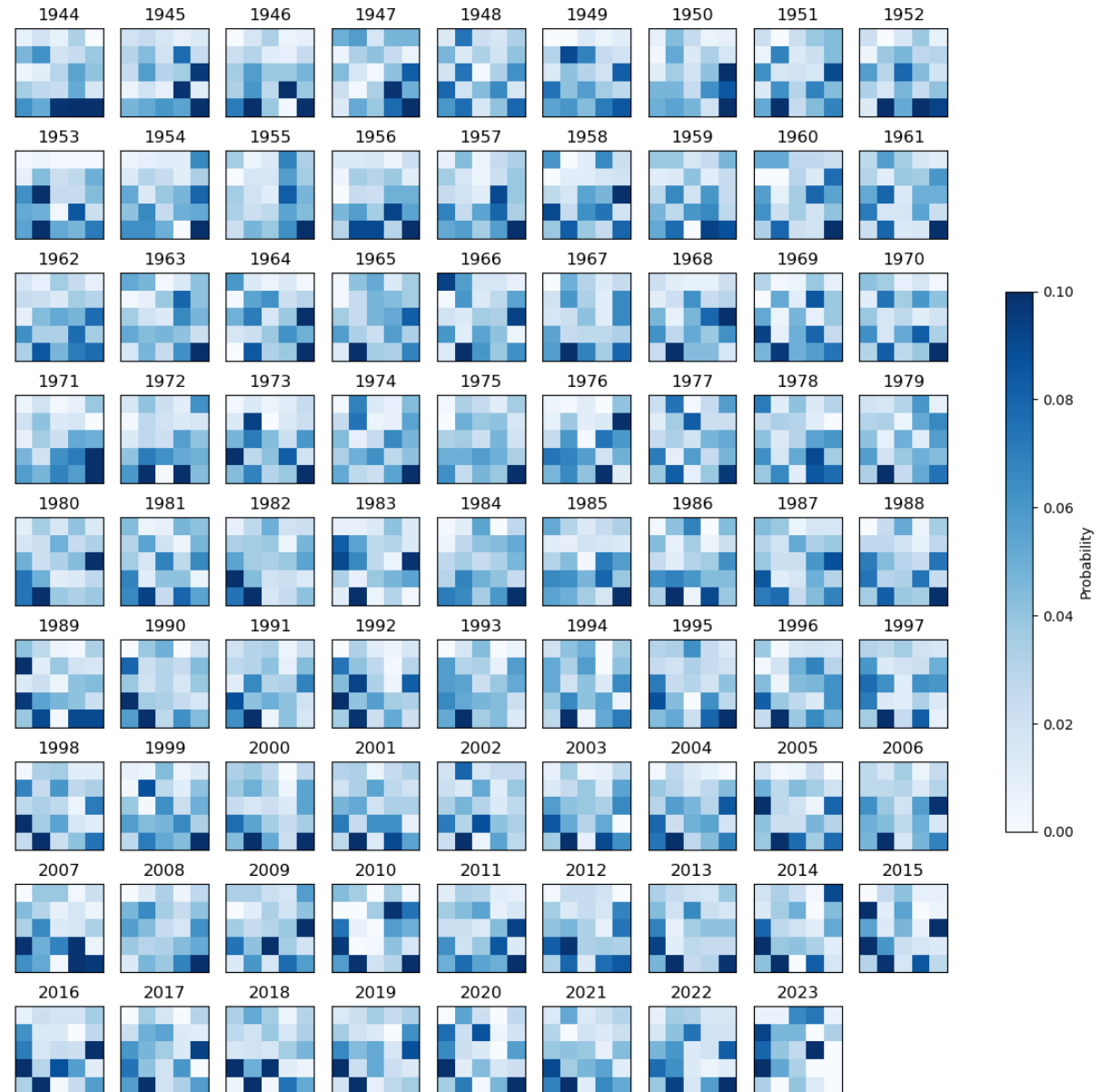
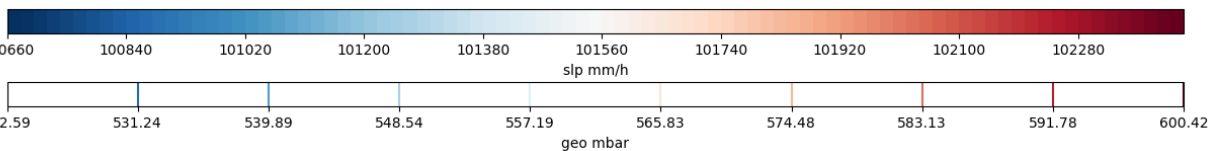
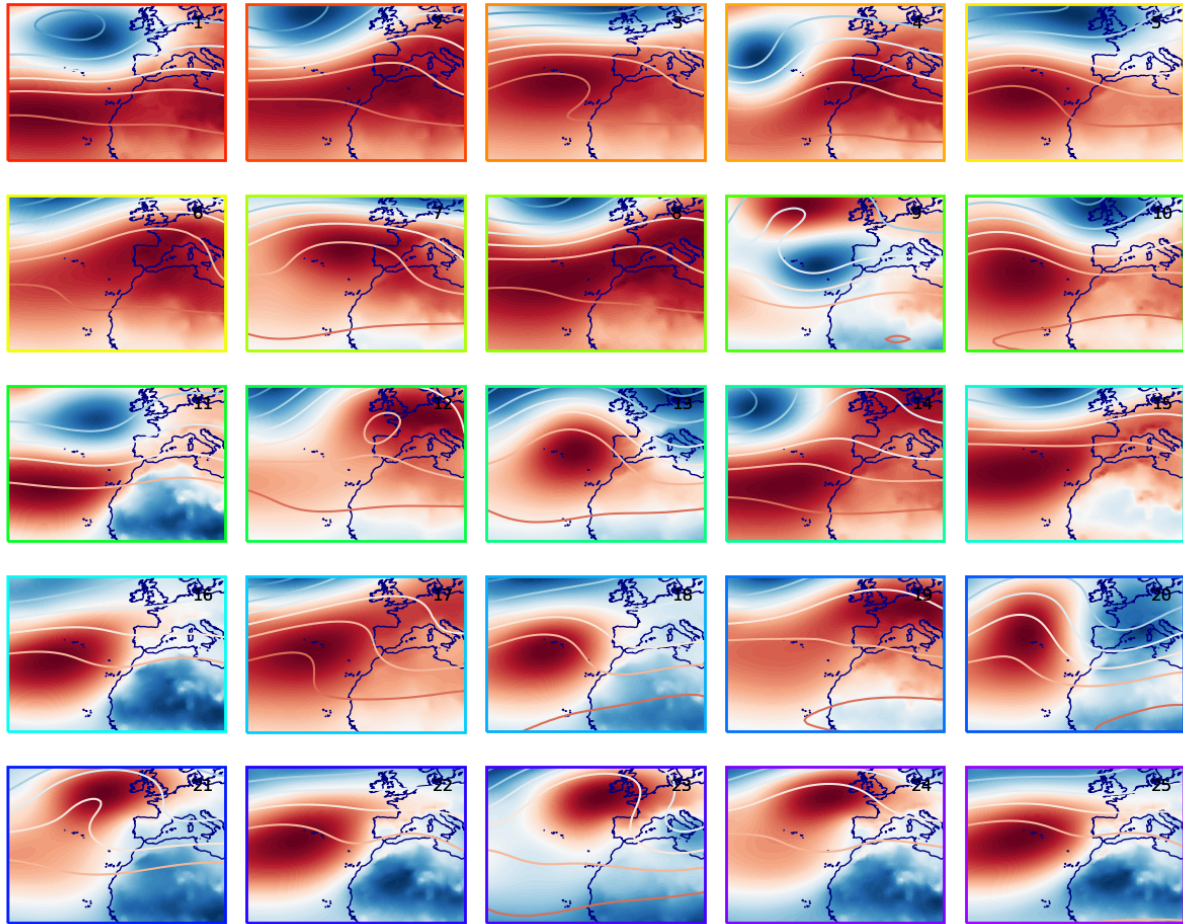


4. AWTs

ANNUAL WEATHER TYPES

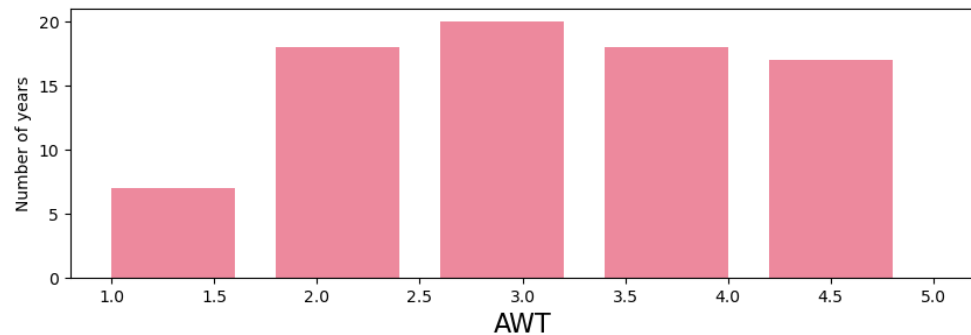
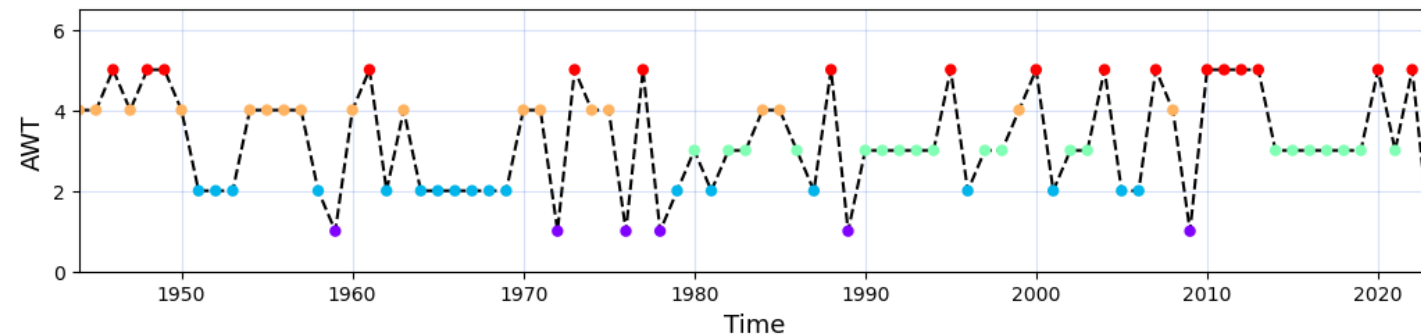
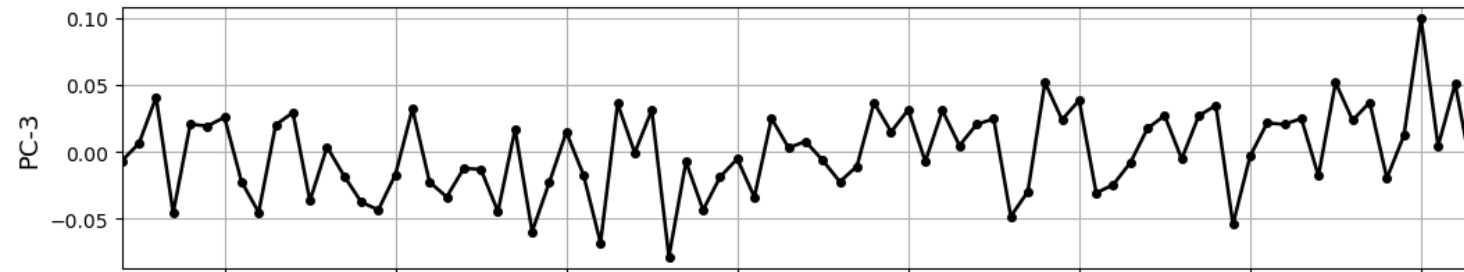
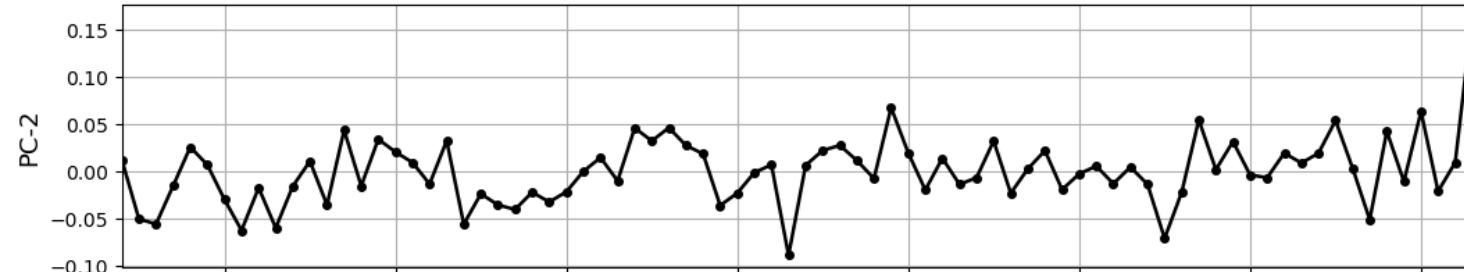
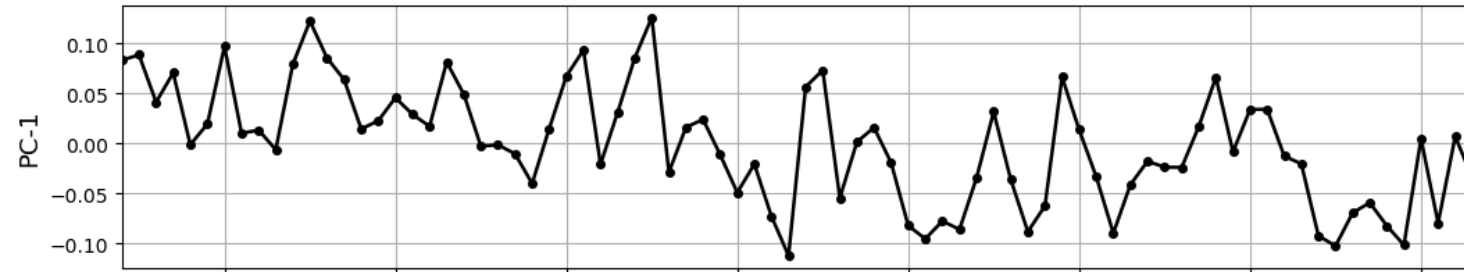
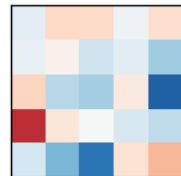
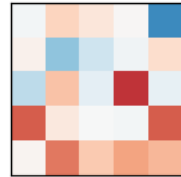
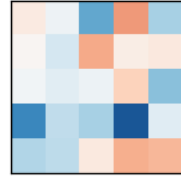


Framework



Framework

Nº PCs = 3
Clusters = 5



Thank you

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<https://geocean.sci.unican.es/hyweb/>



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